Logistic Regression

Logistic Regression

- Logistic Regression is a model for Classification
- Used to model the probability of the class of x via a linear function.
- It can be interpreted as the probability that an input (x) belongs to the default class (Y=1) $[0 \rightarrow 1]$
- The probability of input (x) belongs to class is between 0 and 1

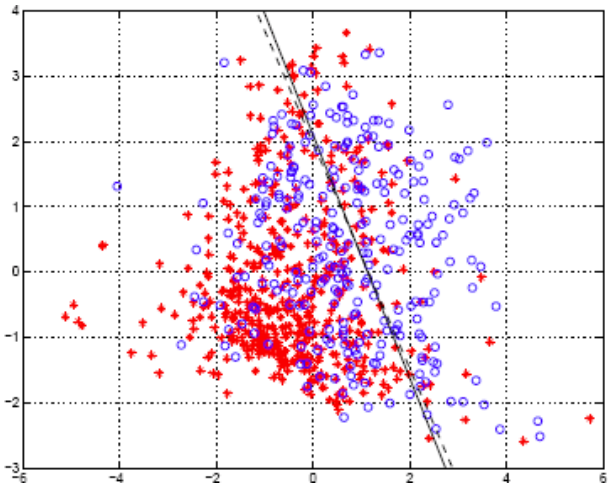
Classification Based on Probability

- Instead of just predicting the class, give the probability of the instance being that class
  - i.e., learn $p(y \mid \mathbf{x})$

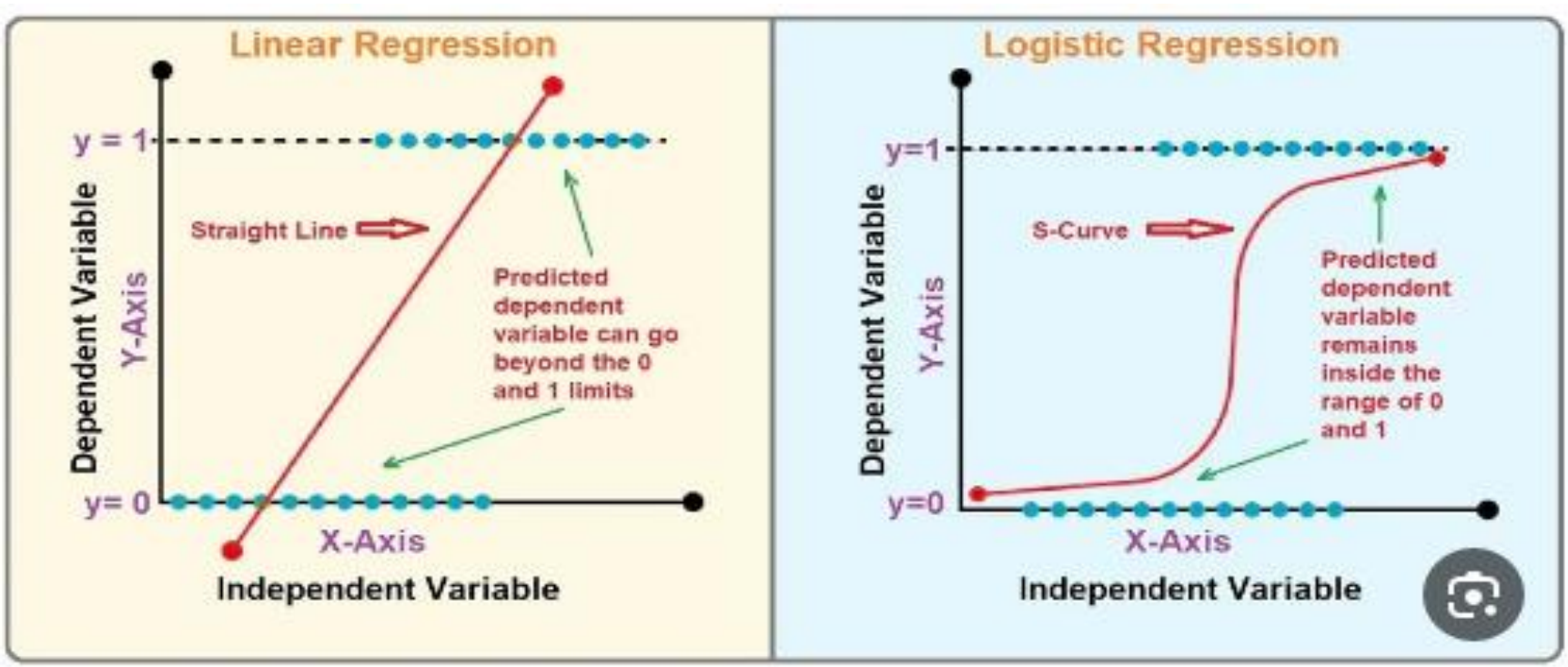
- Recall that:

$$0 \leq p(\text{event}) \leq 1$$

$$p(\text{event}) + p(\neg\text{event}) = 1$$



Logistic Regression



$$h(x) = \theta_0 + \theta_1 x_1$$

If $h(x) \geq 0.5$, Then y=1

If $h(x) < 0.5$, Then y=0

Logistic Regression $0 \leq h(x) \leq 1$

Sigmoid Function $g(z) = \dfrac{1}{1 + e^{-z}}$

Logistic Regression

Probability and Odds

$$p(occurring) = \frac{outcome\ of\ interest}{all\ possible\ outcome}$$

احتمال حدوث

$$odds = \frac{p(occurring)}{p(not\ occurring)}$$

نسبة الاحتمال

$$odds = \frac{p}{1-p}$$

$$p(head) = \frac{1}{2} = 0.5$$

$$odds(heads) = \frac{0.5}{0.5} = 1 \quad or \quad 1:1$$

$$p(1\ or\ 2) = \frac{2}{6} = 0.333$$

$$odds(1\ or\ 2) = \frac{0.333}{0.666} = \frac{1}{2} = 0.5 \quad or\ 1:2$$

$$\ln(odds) = ln\left(\frac{p}{1-p}\right) = logit(p)$$

Logistic Regression

- Takes a probabilistic approach to learning discriminative functions
  - $h_{\boldsymbol{\theta}}(\boldsymbol{x})$ should give $p(y = 1|\boldsymbol{x};\boldsymbol{\theta})$
  - Want $0 \leq h_{\boldsymbol{\theta}}(\boldsymbol{x}) \leq 1$

Can't just use linear regression with a threshold

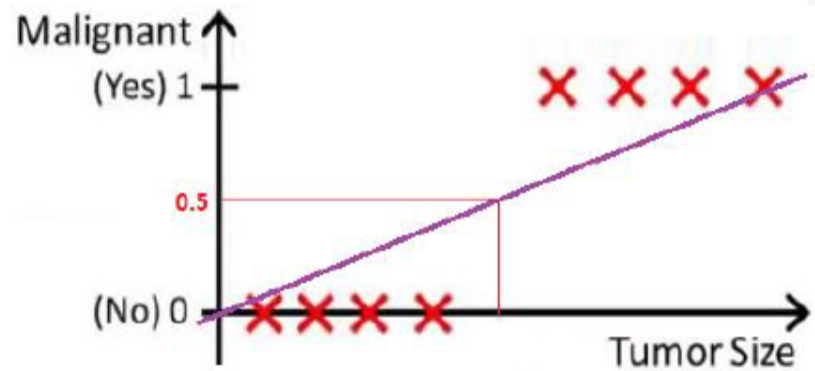
- <u>Odds</u>

  p : probability of an event occurring

  1 – p : probability of the event not occurring

  The odds for event i are then defined as

$$Odds_i = \frac{p_i}{1-p_i}$$



Taking the ***log*** of the odds removes the range restrictions.

$$log_e\left(\frac{p_i}{1-p_i}\right) = \sum_j \theta_j x_{ij} = X_i\theta$$

Logistic Regression

- Takes a probabilistic approach to learning discriminative functions
  - $h_{\boldsymbol{\theta}}(\boldsymbol{x})$ should give $p(y=1|\boldsymbol{x};\boldsymbol{\theta})$
  - Want $0 \leq h_{\boldsymbol{\theta}}(\boldsymbol{x}) \leq 1$

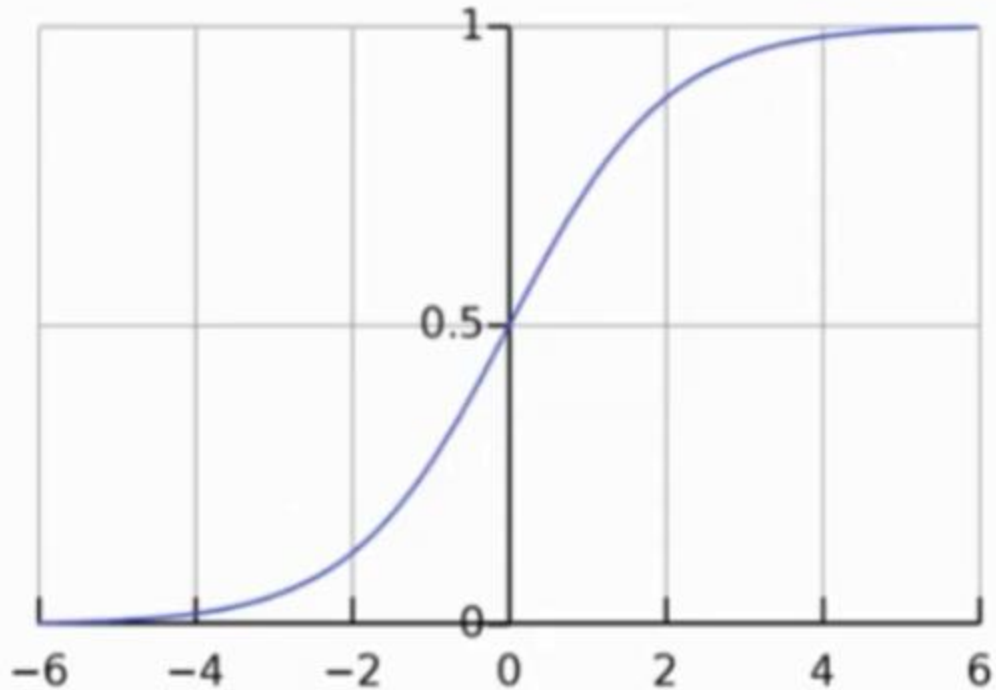
Can't just use linear regression with a threshold

$$log_e\left(\frac{p_i}{1-p_i}\right) = \sum_j \theta_j x_{ij} = X_i\theta$$

$$\frac{p_i}{1-p_i} = e^{X_i\theta}$$

$$p_i = \frac{e^{X_i\theta}}{1+e^{X_i\theta}} = \frac{1}{1+e^{-X_i\theta}} = h_{\boldsymbol{\theta}}(\boldsymbol{x})$$

Logistic Regression



Sigmoid Function $$g(z) = \frac{1}{1 + e^{-z}}$$

$$z = h(x) = \theta_j^T . x_i$$

It's an S-shaped curve that can take any real-valued number and map it into a value between 0 and 1

Logistic Regression

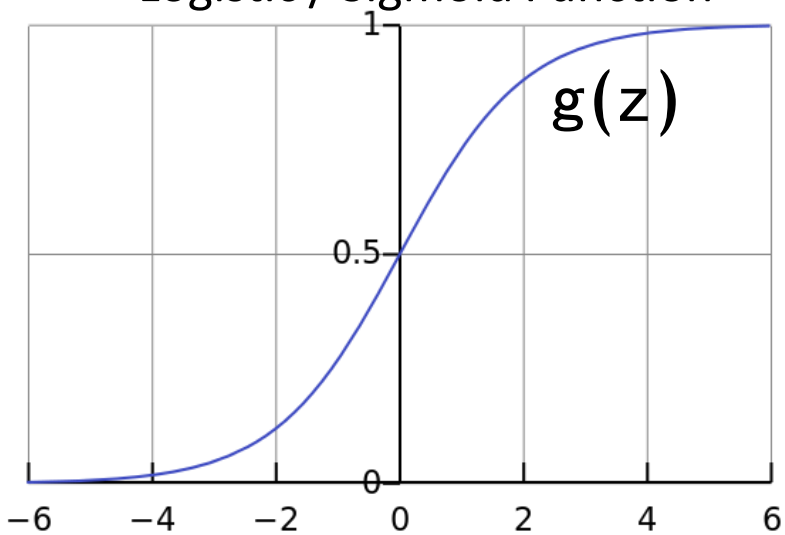
- Logistic regression model:

$$h_{\boldsymbol{\theta}}(\boldsymbol{x}) = g\left(\boldsymbol{\theta}^{\mathsf{T}}\boldsymbol{x}\right)$$

$$g(z) = \frac{1}{1+e^{-z}}$$

$$\boxed{h_{\boldsymbol{\theta}}(\boldsymbol{x}) = \frac{1}{1+e^{-\boldsymbol{\theta}^{\mathsf{T}}\boldsymbol{x}}}}$$

Logistic / Sigmoid Function



Interpretation of Hypothesis Output

$h_{\boldsymbol{\theta}}(\boldsymbol{x})$ = estimated $p(y = 1 \mid \boldsymbol{x}; \boldsymbol{\theta})$

Example: Cancer diagnosis from tumor size

$$\boldsymbol{x} = \begin{bmatrix} x_0 \\ x_1 \end{bmatrix} = \begin{bmatrix} 1 \\ \text{tumorSize} \end{bmatrix}$$

$$h_{\boldsymbol{\theta}}(\boldsymbol{x}) = 0.7$$

→ Tell patient that 70% chance of tumor being malignant

Note that: $p(y = 0 \mid \boldsymbol{x}; \boldsymbol{\theta}) + p(y = 1 \mid \boldsymbol{x}; \boldsymbol{\theta}) = 1$

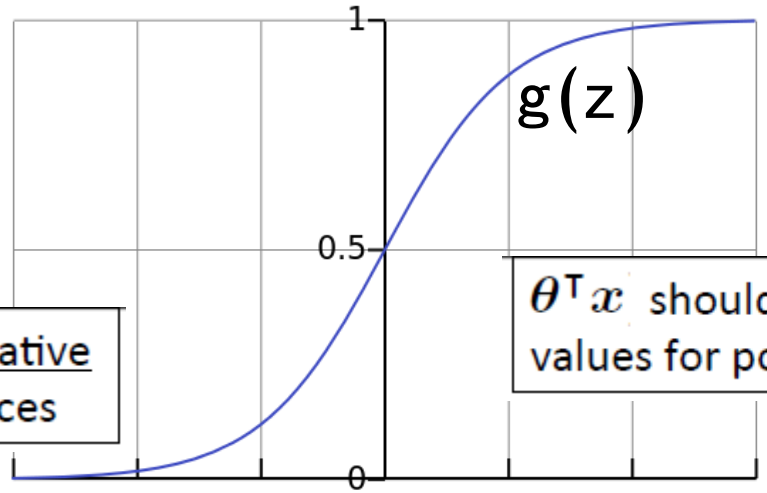
Therefore, $p(y = 0 \mid \boldsymbol{x}; \boldsymbol{\theta}) = 1 - p(y = 1 \mid \boldsymbol{x}; \boldsymbol{\theta})$

Logistic Regression

$$h_{\boldsymbol{\theta}}(\boldsymbol{x}) = g\left(\boldsymbol{\theta}^{\mathsf{T}}\boldsymbol{x}\right)$$

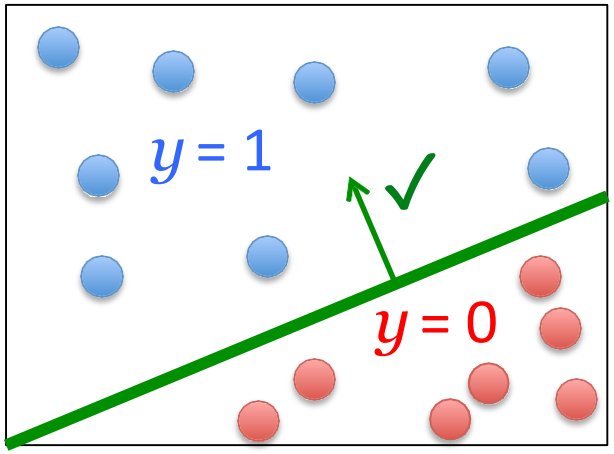
$$g(z) = \frac{1}{1+e^{-z}}$$

$\boldsymbol{\theta}^{\mathsf{T}}\boldsymbol{x}$ should be large <u>negative</u> values for negative instances



$\boldsymbol{\theta}^{\mathsf{T}}\boldsymbol{x}$ should be large <u>positive</u> values for positive instances

- Assume a threshold and...
  - Predict $y = 1$ if $h_{\boldsymbol{\theta}}(\boldsymbol{x}) \geq 0.5$
  - Predict $y = 0$ if $h_{\boldsymbol{\theta}}(\boldsymbol{x}) < 0.5$



Example:

$y \epsilon \{0,1\}$ 0: benign
1: malignant

x: *tumor size*

x	y
3	1
2	1
1	1
5	0
4	0
6	0

$$h(x) = -2x + 6$$

$$z = h(x)$$

$$g(z) = \frac{1}{1 + e^{-(z)}}$$

$$p(y = 1|x;\theta)$$

Estimated probability

$e = 2.718$

x	z	e^{-z}	$1 + e^{-(z)}$	g(z)	y
3	0	e^0 1	2	0.5	1
2	2	$\frac{1}{e^2}$ 0.13536335	1.13536335	0.8808	1
1	4	$\frac{1}{e^4}$ 0.018323237	1.018323237	0.9820	1
5	-4	e^4 54.57551085	55.57551085	0.0179	0
4	-2	e^2 7.387524	8.387524	0.1192	0
6	-6	e^6 403.1778962	404.1778962	0.00247	0

Decision Boundary

$$h(x) = \theta_j^T . x_i$$

$$h(x) = z$$

$$g(z) = \frac{1}{1 + e^{-z}}$$

$$g(z) = p(y = 1|x; \theta)$$

$$p(y = 1|x; \theta) \geq 0.5$$ Threshold

$$when\ z \geq 0$$

$$Then\ p(y = 1|x; \theta) \geq 0.5$$

$$\theta_j^T . x_i \geq 0$$ Decision Boundary

Example

Initially by experience

$$h(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2$$

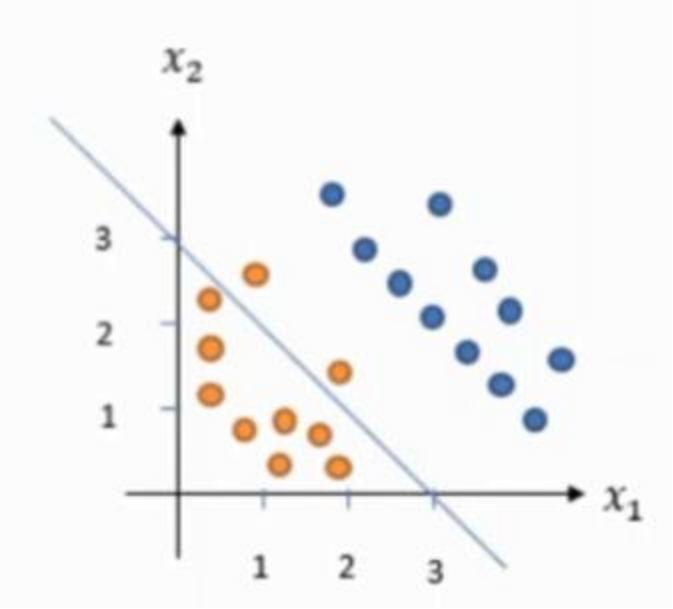
$$\theta_j = \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix}$$

$$h(x) = -3 + x_1 + x_2$$

$$predicty = 1, if - 3 + x_1 + x_2 \geq 0$$

$$x_1 + x_2 \geq 3$$

$$x_1 + x_2 = 3$$

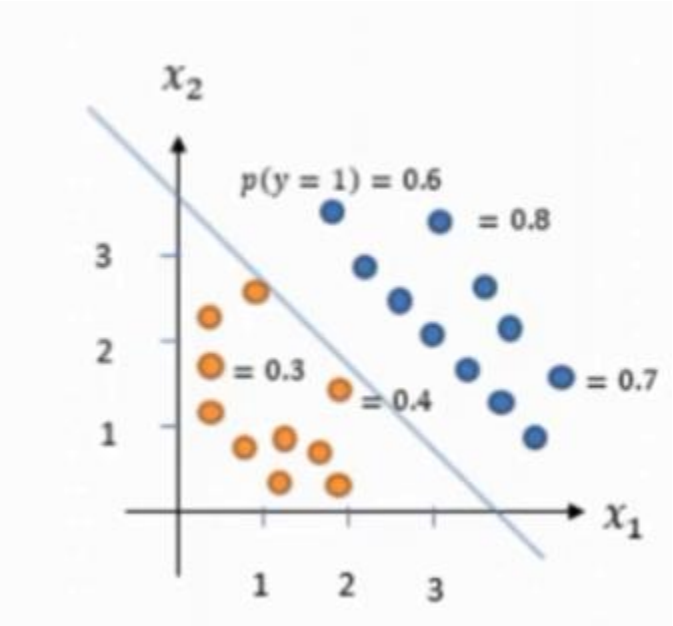


Example

After training

$$\theta_j = \begin{bmatrix} -4 \\ 1 \\ 1 \end{bmatrix}$$

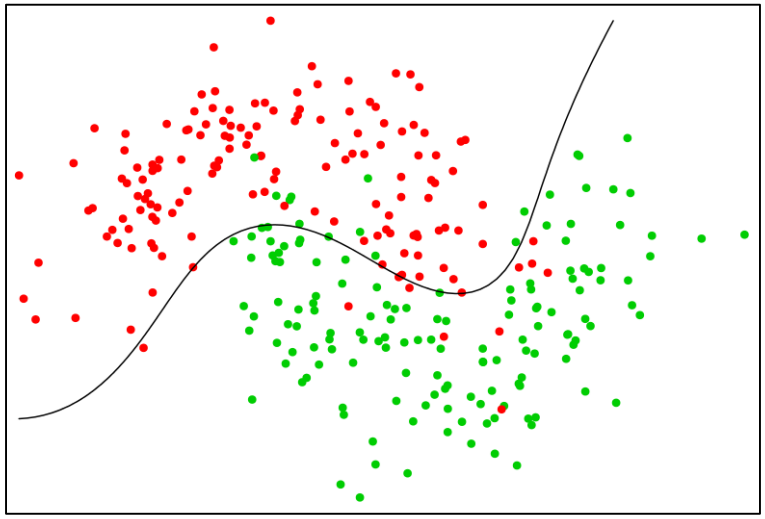
$$h(x) = -4 + x_1 + x_2$$



Non-Linear Decision Boundary

- Can apply basis function expansion to features, same as with linear regression

$$\boldsymbol{x} = \begin{bmatrix} 1 \\ x_1 \\ x_2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 \\ x_1 \\ x_2 \\ x_1 x_2 \\ x_1^2 \\ x_2^2 \\ x_1^2 x_2 \\ x_1 x_2^2 \\ \vdots \end{bmatrix}$$



Non-Linear Decision Boundary

Decision boundary can be:

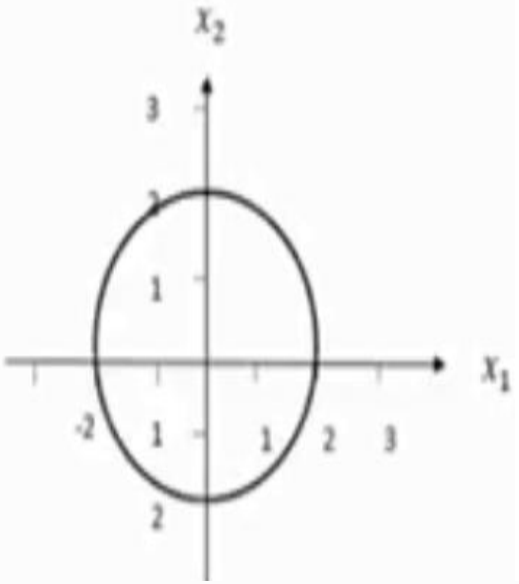
Linear (a line).

Higher order polynomial.

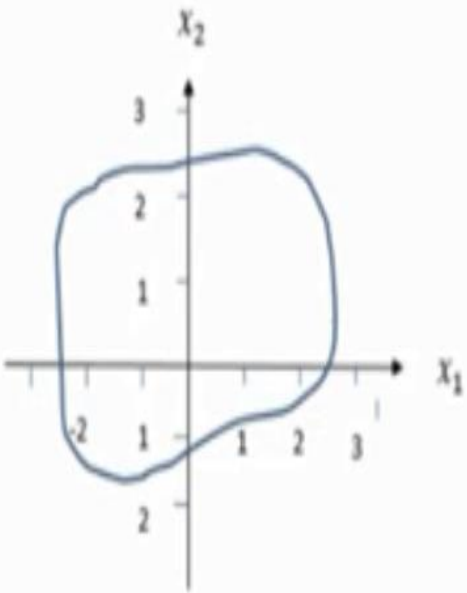
Non-linear.

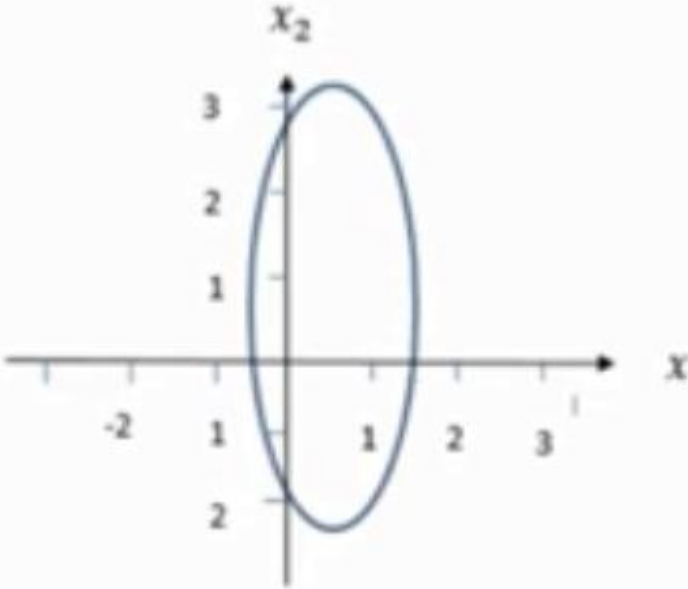
$$h(x) = x_1^2 + x_2^2 - 2$$

$$x_1^2 + x_2^2 \geq 2$$



complex decision boundary.





Logistic Regression

- Given $\left\{\left(\boldsymbol{x}^{(1)}, y^{(1)}\right), \left(\boldsymbol{x}^{(2)}, y^{(2)}\right), \ldots, \left(\boldsymbol{x}^{(n)}, y^{(n)}\right)\right\}$

  where $\boldsymbol{x}^{(i)} \in \mathbb{R}^d, \; y^{(i)} \in \{0, 1\}$

- Model: $h_{\boldsymbol{\theta}}(\boldsymbol{x}) = g\left(\boldsymbol{\theta}^{\mathsf{T}} \boldsymbol{x}\right)$

$$g(z) = \frac{1}{1 + e^{-z}}$$

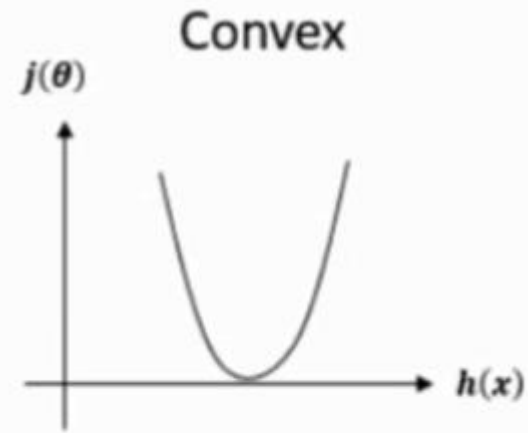
$$\boldsymbol{\theta} = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \vdots \\ \theta_d \end{bmatrix} \qquad \boldsymbol{x}^{\mathsf{T}} = \begin{bmatrix} 1 & x_1 & \ldots & x_d \end{bmatrix}$$

Logistic Regression Objective Function

In Linear regression, to find optimal values for θs:

1. Choose a cost function $$j(\theta) = \frac{1}{2}\sum_{i=1}^{m}(h(x^{(i)}) - y^{(i)})^2$$

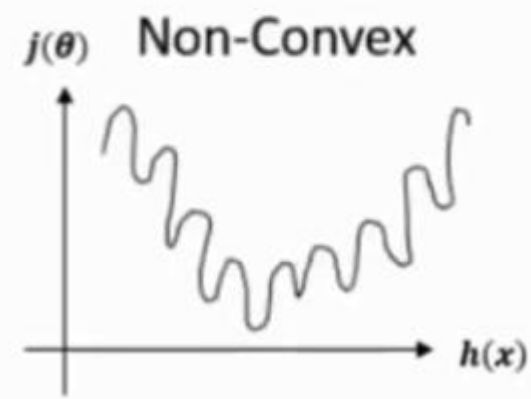
$h(x) = \theta^T x$ is a linear function



What if we choose the same cost function for Logistic regression?

$$g(z) = \frac{1}{1 + e^{-(\theta^T x)}}$$ is a non-linear function

$z = h(x)$



Logistic Regression Objective Function

- Can't just use squared loss as in linear regression:

$$J(\boldsymbol{\theta}) = \frac{1}{2n} \sum_{i=1}^{n} \left(h_{\boldsymbol{\theta}} \left(\boldsymbol{x}^{(i)} \right) - y^{(i)} \right)^2$$

  – Using the logistic regression model

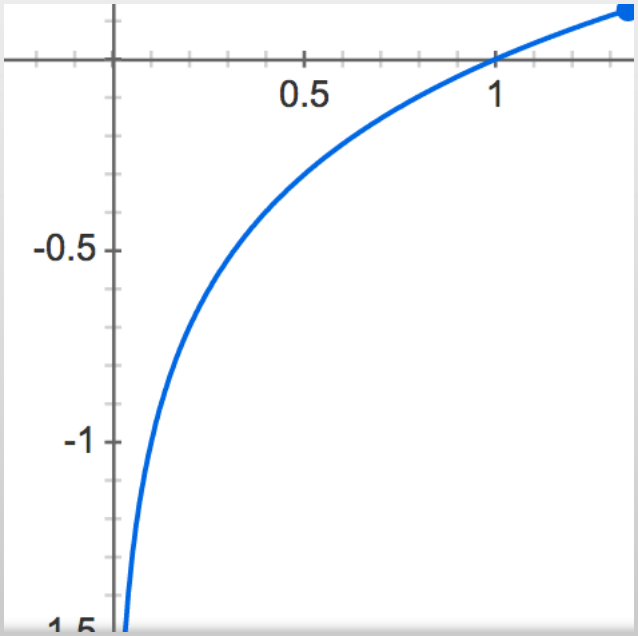
$$h_{\boldsymbol{\theta}}(\boldsymbol{x}) = \frac{1}{1 + e^{-\boldsymbol{\theta}^\top \boldsymbol{x}}}$$

  results in a non-convex optimization

Intuition Behind the Objective

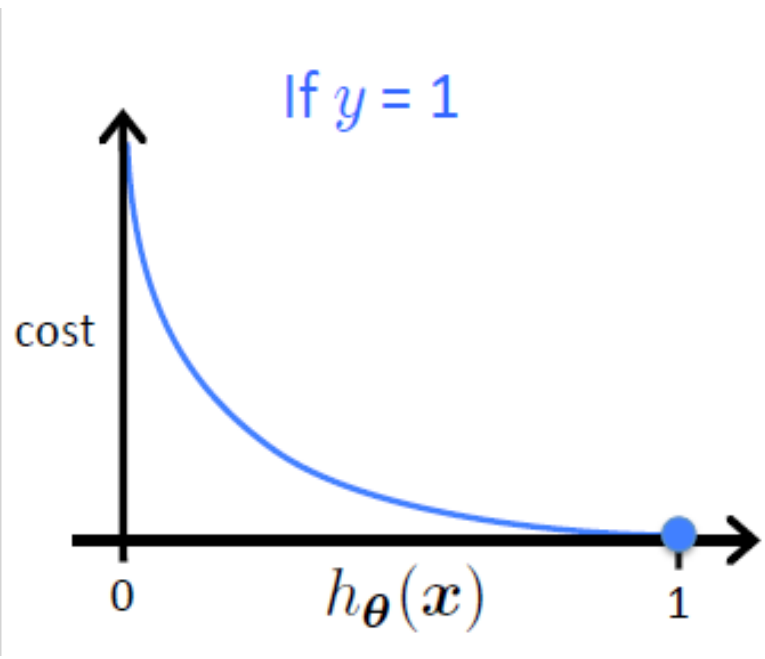
$$\text{cost}\left(h_{\boldsymbol{\theta}}(\boldsymbol{x}), y\right) = \begin{cases} -\log(h_{\boldsymbol{\theta}}(\boldsymbol{x})) & \text{if } y = 1 \\ -\log(1 - h_{\boldsymbol{\theta}}(\boldsymbol{x})) & \text{if } y = 0 \end{cases}$$

Aside: Recall the plot of log(z)



Intuition Behind the Objective

$$\text{cost}\left(h_{\boldsymbol{\theta}}(\boldsymbol{x}), y\right) = \begin{cases} \boxed{-\log(h_{\boldsymbol{\theta}}(\boldsymbol{x})) \quad \text{if } y = 1} \\ -\log(1 - h_{\boldsymbol{\theta}}(\boldsymbol{x})) \quad \text{if } y = 0 \end{cases}$$

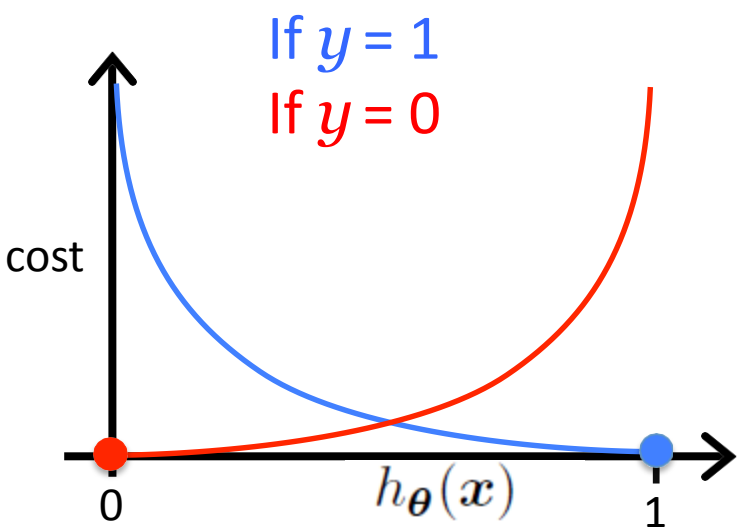


If y = 1

- Cost = 0 if prediction is correct
- As $h_{\boldsymbol{\theta}}(\boldsymbol{x}) \to 0, \text{cost} \to \infty$
- Captures intuition that larger mistakes should get larger penalties
  - e.g., $h_{\boldsymbol{\theta}}(\boldsymbol{x}) = 0$, but y = 1

Intuition Behind the Objective

$$\text{cost}\left(h_{\boldsymbol{\theta}}(\boldsymbol{x}), y\right) = \begin{cases} -\log(h_{\boldsymbol{\theta}}(\boldsymbol{x})) & \text{if } y = 1 \\ -\log(1 - h_{\boldsymbol{\theta}}(\boldsymbol{x})) & \text{if } y = 0 \end{cases}$$

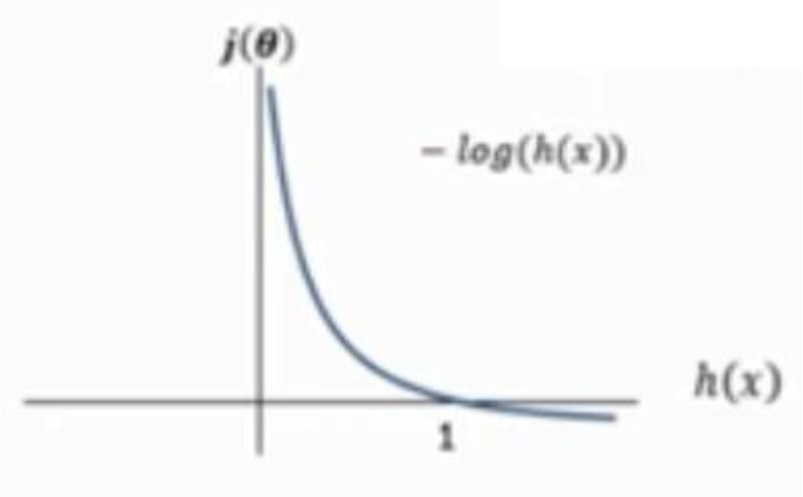


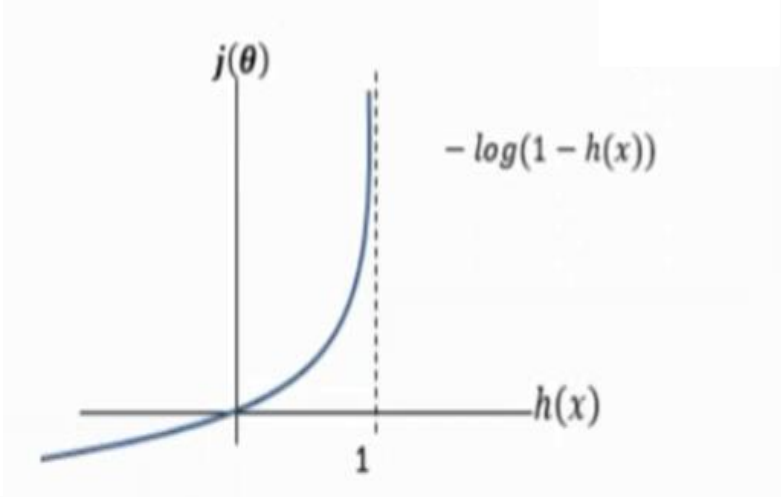
If y = 0

- Cost = 0 if prediction is correct
- As $(1 - h_{\boldsymbol{\theta}}(\boldsymbol{x})) \to 0, \text{cost} \to \infty$

As Captures intuition that larger mistakes should get larger penalties

Intuition Behind the Objective





Actual y	Predicted h(x)	$j(\theta)$
1	0.9	0.105
1	0.5	0.693
1	0.1	2.302
1	0.01	4.605

Actual y	Predicted h(x)	$1-h(x)$	$j(\theta)$
0	0.9	0.1	2.302
0	0.5	0.5	0.693
0	0.1	0.9	0.105
0	0.01	0.99	0.01

Deriving the Cost Function via Maximum Likelihood Estimation

Logistic regression objective:

$$\min_{\boldsymbol{\theta}} J(\boldsymbol{\theta})$$

$$J(\boldsymbol{\theta}) = -\sum_{i=1}^{n} \left[y^{(i)} \log h_{\boldsymbol{\theta}}(\boldsymbol{x}^{(i)}) + \left(1 - y^{(i)}\right) \log \left(1 - h_{\boldsymbol{\theta}}(\boldsymbol{x}^{(i)})\right) \right]$$

Intuition Behind the Objective

$$J(\boldsymbol{\theta}) = -\sum_{i=1}^{n}\left[y^{(i)}\log h_{\boldsymbol{\theta}}(\boldsymbol{x}^{(i)}) + \left(1-y^{(i)}\right)\log\left(1-h_{\boldsymbol{\theta}}(\boldsymbol{x}^{(i)})\right)\right]$$

- Cost of a single instance:

$$\text{cost}\left(h_{\boldsymbol{\theta}}(\boldsymbol{x}), y\right) = \begin{cases} -\log(h_{\boldsymbol{\theta}}(\boldsymbol{x})) & \text{if } y=1 \\ -\log(1-h_{\boldsymbol{\theta}}(\boldsymbol{x})) & \text{if } y=0 \end{cases}$$

- Can re-write objective function as

$$J(\boldsymbol{\theta}) = \sum_{i=1}^{n}\text{cost}\left(h_{\boldsymbol{\theta}}(\boldsymbol{x}^{(i)}), y^{(i)}\right)$$

Compare to linear regression: $J(\boldsymbol{\theta}) = \frac{1}{2n}\sum_{i=1}^{n}\left(h_{\boldsymbol{\theta}}\left(\boldsymbol{x}^{(i)}\right) - y^{(i)}\right)^2$

Regularized Logistic Regression

$$J(\boldsymbol{\theta}) = -\sum_{i=1}^{n}\left[y^{(i)}\log h_{\boldsymbol{\theta}}(\boldsymbol{x}^{(i)}) + \left(1 - y^{(i)}\right)\log\left(1 - h_{\boldsymbol{\theta}}(\boldsymbol{x}^{(i)})\right)\right]$$

- We can regularize logistic regression exactly as before:

$$\begin{aligned} J_{\text{regularized}}(\boldsymbol{\theta}) &= J(\boldsymbol{\theta}) + \frac{\lambda}{2}\sum_{j=1}^{d}\theta_j^2 \\ &= J(\boldsymbol{\theta}) + \frac{\lambda}{2}\|\boldsymbol{\theta}_{[1:d]}\|_2^2 \end{aligned}$$

Gradient Descent for Logistic Regression

$$J_{\text{reg}}(\boldsymbol{\theta}) = -\sum_{i=1}^{n}\left[y^{(i)}\log h_{\boldsymbol{\theta}}(\boldsymbol{x}^{(i)}) + \left(1 - y^{(i)}\right)\log\left(1 - h_{\boldsymbol{\theta}}(\boldsymbol{x}^{(i)})\right)\right] + \frac{\lambda}{2}\|\boldsymbol{\theta}_{[1:d]}\|_2^2$$

Want $\min_{\boldsymbol{\theta}} J(\boldsymbol{\theta})$

- Initialize $\boldsymbol{\theta}$
- Repeat until convergence

$$\theta_j \leftarrow \theta_j - \alpha\frac{\partial}{\partial\theta_j}J(\boldsymbol{\theta})$$

simultaneous update for j = 0 ... d

Use the natural logarithm (ln = $\log_e$) to cancel with the exp() in $h_{\boldsymbol{\theta}}(\boldsymbol{x})$

Gradient Descent for Logistic Regression

$$J_{\text{reg}}(\boldsymbol{\theta}) = -\sum_{i=1}^{n}\left[y^{(i)}\log h_{\boldsymbol{\theta}}(\boldsymbol{x}^{(i)}) + \left(1-y^{(i)}\right)\log\left(1-h_{\boldsymbol{\theta}}(\boldsymbol{x}^{(i)})\right)\right] + \frac{\lambda}{2}\|\boldsymbol{\theta}_{[1:d]}\|_2^2$$

Want $\min_{\boldsymbol{\theta}} J(\boldsymbol{\theta})$

- Initialize $\boldsymbol{\theta}$
- Repeat until convergence (simultaneous update for $j = 0 \ldots d$)

$$\theta_0 \leftarrow \theta_0 - \alpha \sum_{i=1}^{n}\left(h_{\boldsymbol{\theta}}\left(\boldsymbol{x}^{(i)}\right) - y^{(i)}\right)$$

$$\theta_j \leftarrow \theta_j - \alpha\left[\sum_{i=1}^{n}\left(h_{\boldsymbol{\theta}}\left(\boldsymbol{x}^{(i)}\right) - y^{(i)}\right)x_j^{(i)} + \lambda\theta_j\right]$$

Gradient Descent for Logistic Regression

- Initialize $\boldsymbol{\theta}$
- Repeat until convergence (simultaneous update for $j = 0 \ldots d$)

$$\theta_0 \leftarrow \theta_0 - \alpha \sum_{i=1}^{n} \left(h_{\boldsymbol{\theta}} \left(\boldsymbol{x}^{(i)} \right) - y^{(i)} \right)$$

$$\theta_j \leftarrow \theta_j - \alpha \left[\sum_{i=1}^{n} \left(h_{\boldsymbol{\theta}} \left(\boldsymbol{x}^{(i)} \right) - y^{(i)} \right) x_j^{(i)} + \lambda \theta_j \right]$$

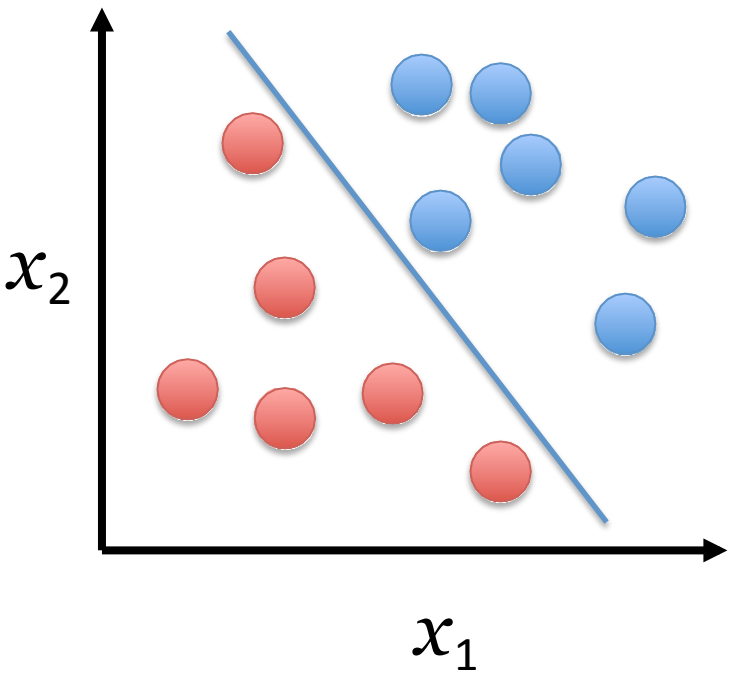
This looks IDENTICAL to linear regression!!!

- Ignoring the $1/n$ constant
- However, the form of the model is very different:

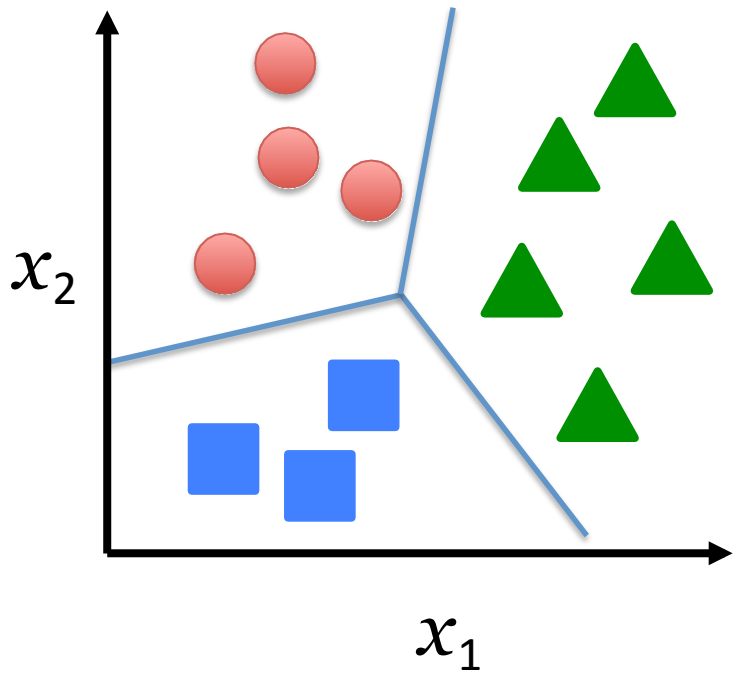
$$h_{\boldsymbol{\theta}}(\boldsymbol{x}) = \frac{1}{1 + e^{-\boldsymbol{\theta}^{\mathsf{T}} \boldsymbol{x}}}$$

Multi-Class Classification

Binary classification:



Multi-class classification:

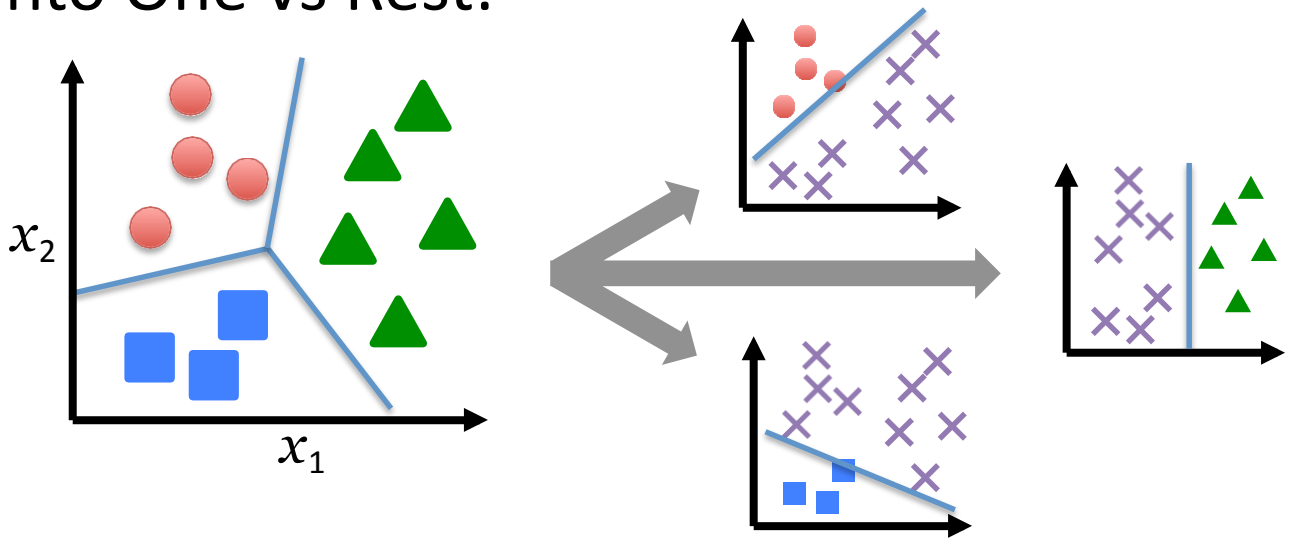


Disease diagnosis: healthy / cold / flu / pneumonia

Object classification: desk / chair / monitor / bookcase

Multi-Class Logistic Regression

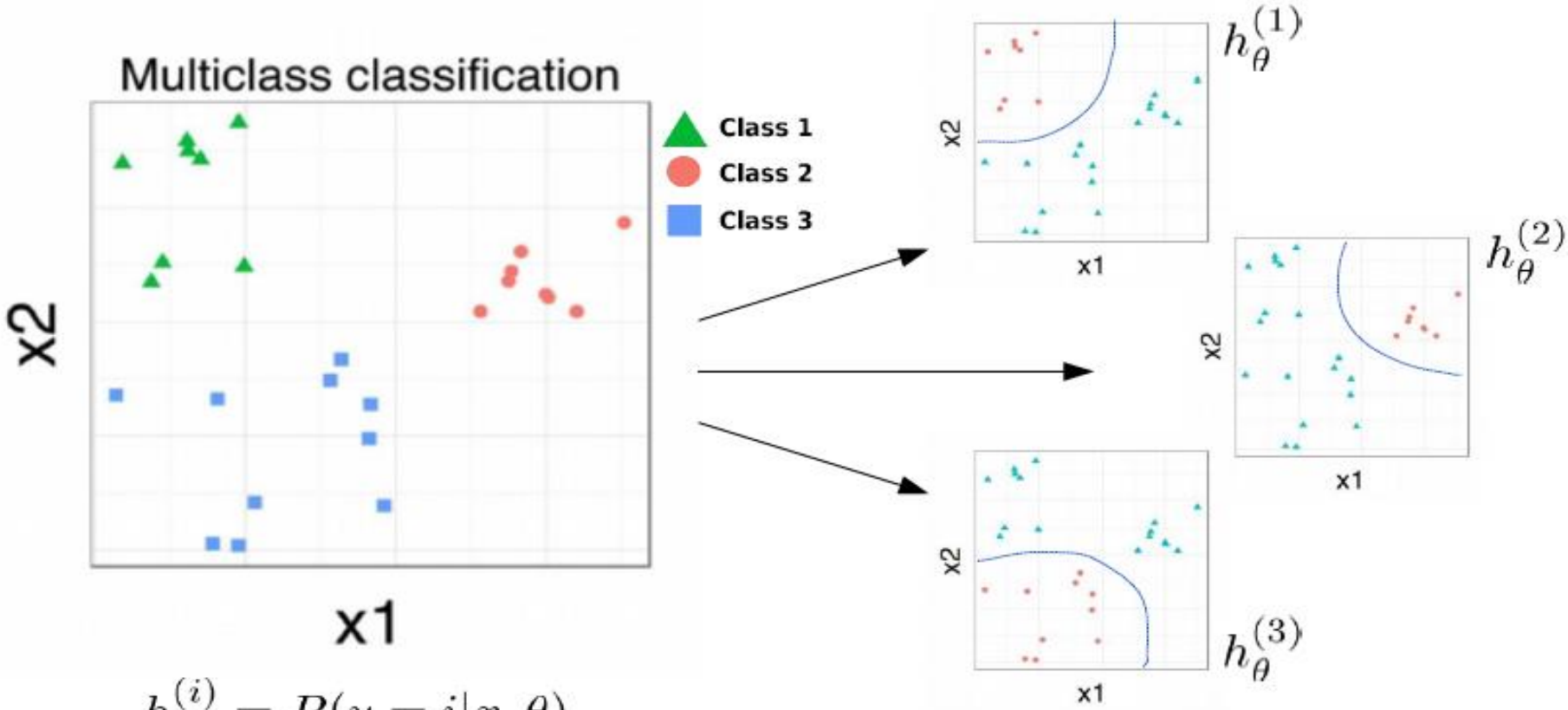
Split into One vs Rest:



- Predict class label as the most probable label

Multi-Class Logistic Regression

One versus all (one-vs-rest)



$$h_\theta^{(i)} = P(y = i | x, \theta)$$

where $i \in \{1, 2, 3\}$

Multi-Class Logistic Regression

- **Train logistic regression classifier** $h_\theta^{(i)}(x)$ **for each class** *i* **to predict probability of** *y* = *i*
- **On new** *x* **predict class** *i* **which satisfies**

$$\underset{i}{argmax}\ h_\theta^{(i)}(x)$$